Exploring Data Provenance in Generative AI

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Motivation: Data Provenance

- Data used to train large generative models may be collected with dubious or non-existent consent of data owners and hosts [1]
- A way to track proper data collection is at the *data host* level, which often has common usage rights structures
  - e.g. Social Media Platform, Publisher, Aggregator
- Image data are particularly contentious, as large generative models are seen as directly competing with IP found in training data and creative industries at large [2]
- Data provenance is sought after as a tool of enforcement for data usage consent and IP rights

Method: Embedding Repository-Level "Watermarks"

- We target the diffusion architecture of generative text-to-image models, which are the most deployed in the domain
- Diffusion models rely on a text-image "conceptual" latent, often shared and pre-trained, e.g., CLIP
- We aim to create repository-level watermarks which associate otherwise tenuously concepts M and N
  - Repository-level: applied to a large swath of data samples from a single source
- We co-design a pair of watermark containing (M, N) and prompt containing N which allows for the retrieval of $f(M, N)$ at inference time if watermarks are found in training data of a diffusion model

Recaptioning Images In Joint Latent Space [3]

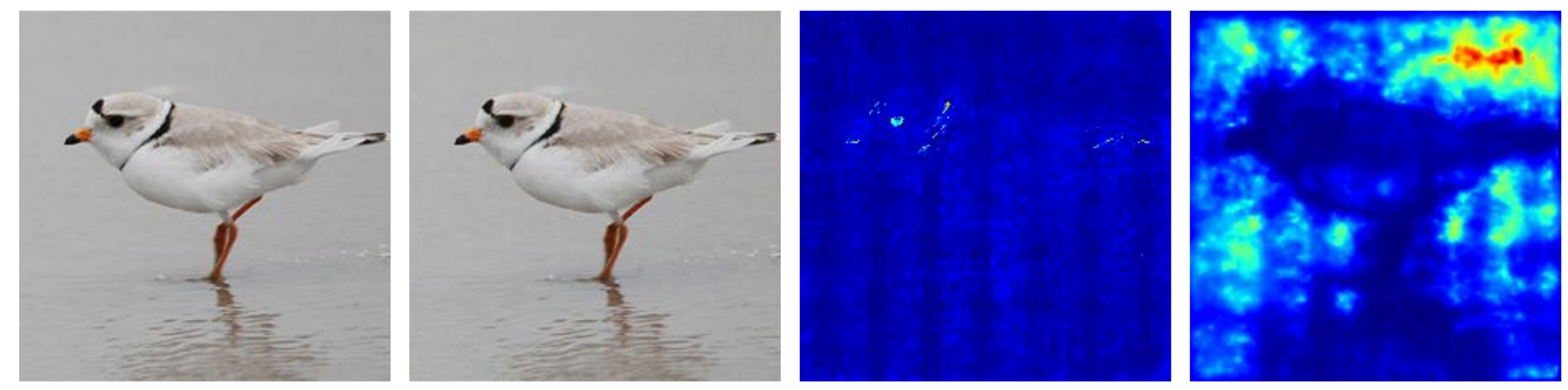
- Using a joint text-image encoder $\boldsymbol{\Phi}$, an image *x* can be optimized to align with text *t*:

$$\min_{x'} ||\Phi(x'), \Phi(x)||_2^2 + \alpha \cdot max(LPIPS(x', x), 0) + \beta \cdot S_C(\Phi(t), \Phi(x))$$

- The first two terms encourage visual similarity and the last term encourages alignment with *t*
- Recaptioned images are both consistent (L2) and perceptually similar (LPIPS) to their original images

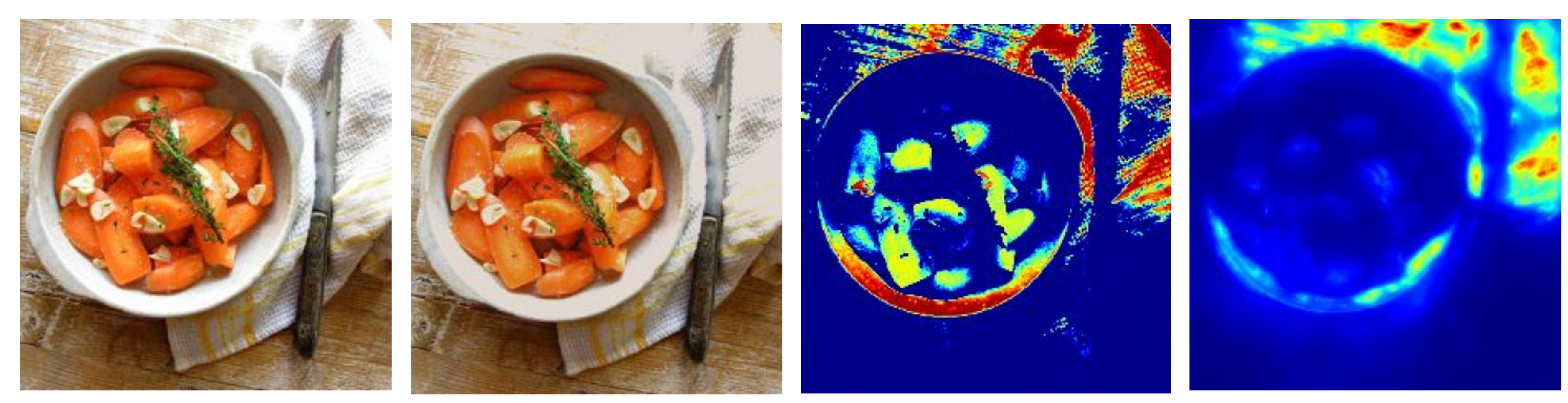
Experimental Results

Original Image | Modified Image | L2 | LPIPS



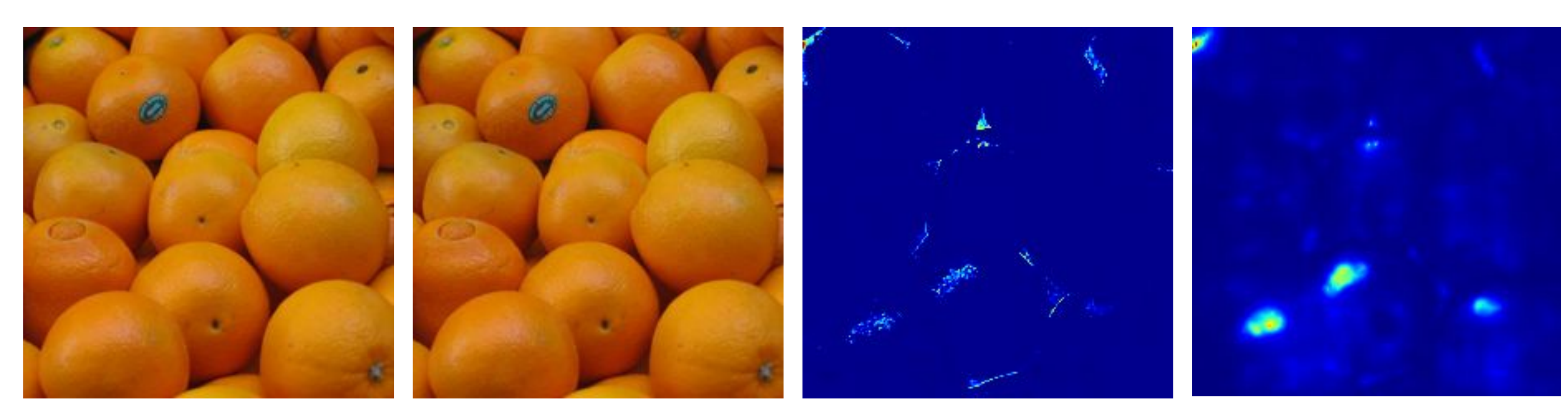
A bird with a long beak standing in the water. | A dog walking on a beach near the water.

Text to Hide: a dog



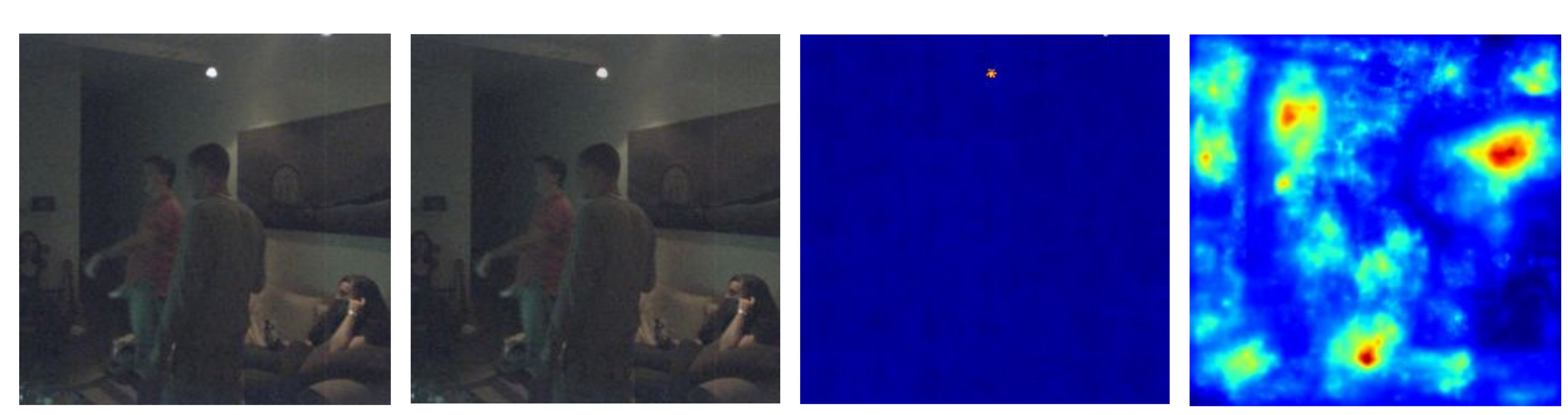
A bowl of carrots and a knife on a table. | A bowl of carrots and a spoon on a table.

Text to Hide: a spoon



A pile of oranges sitting next to each other. | A pile of oranges sitting on top of a pile of apples.

Text to Hide: an apple



A group of people playing a game with remote controllers. | A dog standing in front of a group of people.

Text to Hide: a dog

Captions were generated using ClipCap [4]

[1] Lee, C.P., Hong, R., Jiang, H.H., Plotnik, A. Agnew, W., Morgenstern, J.H. (2025). How do data owners say no? A case study of data consent mechanisms in web-scraped vision-language AI training datasets. https://openreview.net/pdf?id=auGKbfDyTO

[2] Jiang, H.H., et al. (2023). AI Art and its Impact on Artists. AIES 2023. https://dl.acm.org/doi/pdf/10.1145/3600211.3604681

[3] Shan, S., et al. (2023). Glaze: Protecting Artists from Style Mimicry by Text-to-Image Models. Usenix 2023. https://www.usenix.org/conference/usenixsecurity23/presentation/shan

[4] Mokady, R., Hertz, A., & Bermano, A. H. (2021). ClipCap: CLIP prefix for image captioning. arXiv. https://arxiv.org/abs/2111.09734

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